



Mirror cooled to quantum state

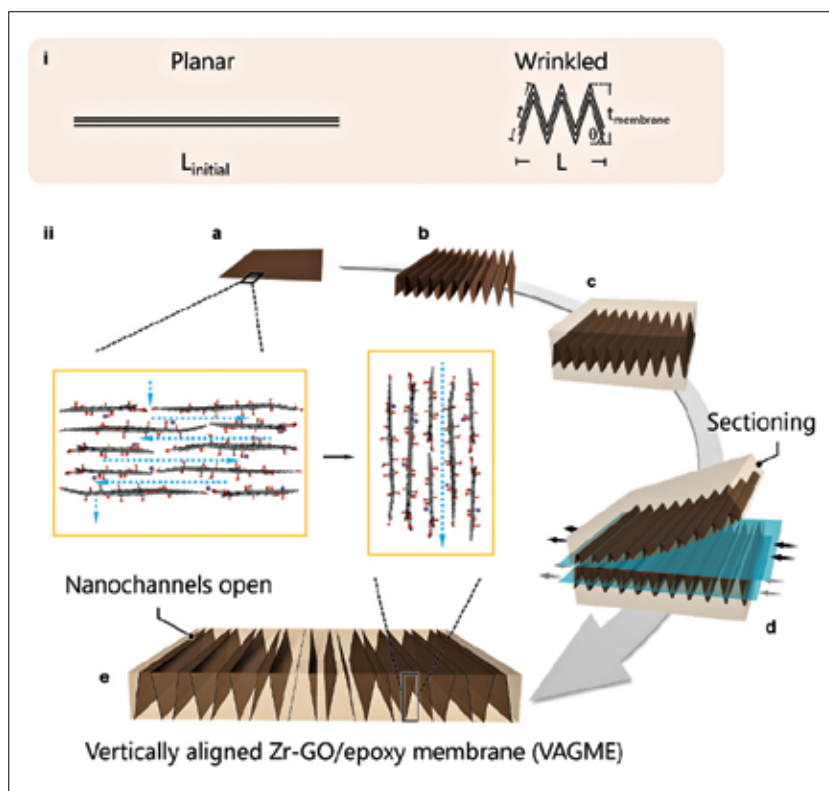
Researchers have cooled a 40kg LIGO mirror to near its quantum ground state. <https://digit.in/jul21-07>



Quantum network created

Researchers have successfully sent quantum information over 600-km-long optical fibers. <https://digit.in/jul21-08>

coffee filter, where not all the gaps are the same size, with clusters of larger or smaller holes. Within the desalination membranes, in three dimensions, a similar thing is taking place, where water finds it tougher to pass through some areas and not the others. The polymer films used in the membranes were mapped at a spatial resolution of about one nanometer, which is less than half a diameter of a DNA strand. These density and mass variations on a nanoscale are invisible to the naked eye, but the researchers found that these play a major role in the efficiency of the water filters. The researchers found that it was not the thickness of the membrane itself that mattered so much, but rather the concentration and number of “dead zones”, highly dense regions in the membrane that made it very difficult for water to pass through. The researchers estimate that making densities more uniform can increase membrane efficiency by as much as 30 to 40 percent, resulting in more water being filtered with less energy use. Enrique Gomez, professor of chemical engineering and materials science and engineering at Penn State and lead author of the paper said, “Freshwater management is becoming a crucial challenge throughout the world. Shortages, droughts — with increasing severe weather patterns, it is expected this problem will become even more significant. It’s critically important to

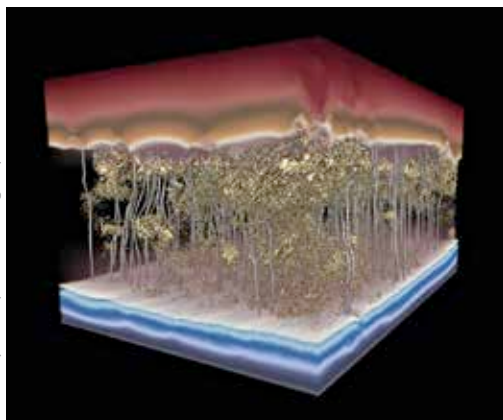


The wrinkled graphene nanochannels providing superior filtration

have clean water available, especially in low resource areas.” The researchers are now investigating approaches to make the best membranes out of specific materials, such as those with antimicrobial properties.

On the smallest scales, it is not only the materials that make a difference but also the spaces between them. By the precise stacking of two dimensional nanomaterials such as graphene, it is possible to use the gaps in different ways. Researchers have explored using these gaps for storage, growing cells and the field of nanofluidics has using narrow channels to selectively allow some molecules to pass through. These gaps are known as nanochannels, and researchers from Brown University have found a way of using these nanochannels for the specific purpose of filtering water. When graphene sheets are stacked for filtering water, the sheets are

parallel to each other, and so are the nanochannels, which means that the water has to travel through a lot of sheets to get adequately filtered. For the water to travel a shorter distance and make smaller filters, it was needed for the nanochannels and the graphene sheets to be oriented vertically. There was no good way to do this, till now. The new approach developed deposits graphene sheets on an elastic substrate, which crumples up after the graphene is deposited, leading to stacked sheets that are almost vertical. The graphene sheets within the wrinkled up elastic substrate form very short and narrow channels for the water to pass through. The nanochannels align themselves almost vertically, just like the graphene sheets, after which they are encased in epoxy. The graphene combined with the nanochannels allow only small molecules to pass through, trapping even such contaminants as metal ions. The assemblages are dubbed VAGMEs (vertically aligned graphene membranes). 



The density variation in filtration membranes can affect how much water is produced